THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 5 & 6: **Life Processes, Control and Coordination** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of these is not required for photosynthesis? 1

(a) carbon dioxide **(b)** chlorophyll **(c)** oxygen **(d)** sunlight

2. Trypsin acts on: 1

(a) starch **(b)** proteins **(c)** nucleic acids **(d)** fats

3. Exchange of gases in lungs takes place in: 1

(a) trachea **(b)** larynx **(c)** alveoli **(d)** bronchi

4. A student states: “Veins have valves to prevent backflow of blood.” The correct response is: 1

(a) Correct **(b)** Incorrect

(c) Only arteries have valves **(d)** Cannot be determined

5. The blood vessel with the thickest wall is the: 1

(a) capillary **(b)** vein **(c)** lymph vessel **(d)** artery

6. The longest fibre of a nerve cell is the: 1

(a) axon **(b)** cell body **(c)** dendrite **(d)** synapse

7. The part of the brain associated with thinking is: 1

(a) cerebellum **(b)** medulla **(c)** cerebrum **(d)** pons

8. Insulin is secreted by: 1

(a) adrenal **(b)** pancreas **(c)** thyroid **(d)** liver

9. A student states: “Plants have a nervous system.” The correct response is: 1

(a) Correct

(b) Only flowering plants

(c) Incorrect — plants use hormones and cell changes

(d) Cannot be determined

10. Which hormone regulates carbohydrate, protein and fat metabolism? 1

(a) testosterone **(b)** thyroxine **(c)** adrenaline **(d)** oestrogen

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. What are the differences between autotrophic and heterotrophic nutrition? 2

12. Name the two types of cells in which anaerobic respiration occurs. 2

13. Why does a person suffering from diabetes need insulin injections? 2

14. Name two tissues that respond to nerve impulses. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. What are the three events during photosynthesis? 3

16. How is the brain protected? Why is it necessary? 3

17. Describe an experiment to show that CO_2 is essential for photosynthesis. 3

OR

Explain how oxygen and carbon dioxide are transported in human beings. 3

18. What is the difference between a reflex action and walking? 3

OR

Explain how plant growth is influenced by hormones (auxin, gibberellin, cytokinin). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) What is nutrition?
(ii) Describe the steps of holozoic nutrition in Amoeba with a diagram.
(iii) How is it different from that of Paramecium? 5

OR

Case study: Activity: freshly prepared lime water in two test tubes; air is pumped into one and exhaled air is blown into the other. 5
(i) In which tube does lime water turn milky faster?
(ii) Why?
(iii) What is the gas responsible?
(iv) What is the composition difference between inhaled and exhaled air?
(v) Name the process in which energy is released from food.

20. (i) Draw a neat diagram (describe) of a neuron.
(ii) Explain how an impulse travels from one neuron to the next.
(iii) What are sensory and motor neurons? 5

OR

Case study: Ravi touched a hot iron and pulled his hand away instantly before he felt pain. 5
(i) Name this action.
(ii) Which part of the nervous system controls it?
(iii) Draw a flow chart of the path.
(iv) Why did he feel pain later?
(v) Give one more example.

END OF QUESTION PAPER — **Total: 40 Marks**